$\boxed{N5}$

$[\theta; 2\theta]$

$\xi \sim R(\theta, 2\theta)$

$p(x, \theta) = \frac{1}{\theta} \{(\theta, 2\theta)\}$

$$\alpha_k = \int_{-\infty}^{\infty} x^k \, df(x, \theta) = \alpha_k(\theta)$$

$$M\xi = \alpha_1 = \int_{\theta}^{2\theta} x \cdot \frac{1}{\theta} dx = \ldots = \frac{3}{2}\theta$$

$$M\xi^2 = \alpha_2 = \int_{\theta}^{2\theta} \frac{x^2}{\theta} dx = \frac{7}{3}\theta^2$$

$$\mu_2 = D\xi = M(\xi^2) - (M\xi)^2 = \frac{\theta^2}{12}$$

<u>ОММ:</u> $\alpha_1(\theta) = \hat{\alpha}_1$

$$\frac{3}{2}\theta = \bar{x} \Rightarrow \hat{\theta}_1 = \frac{2}{3}\bar{x}$$

Несм.: $M\hat{\theta}_1 = \theta$:

$$M\hat{\theta}_1 = M\left(\frac{2}{3}\bar{x}\right) =$$

$$= \frac{2}{3} M\xi = \frac{2}{3} \cdot \frac{3}{2}\theta = \theta$$

$\Rightarrow \hat{\theta}_1$ – несмещ.

Сост.: $D\hat{\theta}_1 = \frac{4}{9n} D\xi = \frac{4\theta^2}{108n} =$

$$= \frac{\theta^2}{27n} \xrightarrow[n\to\infty]{} 0 \xrightarrow{\text{дост. усл.}} \hat{\theta}_1 \text{ - сост.}$$

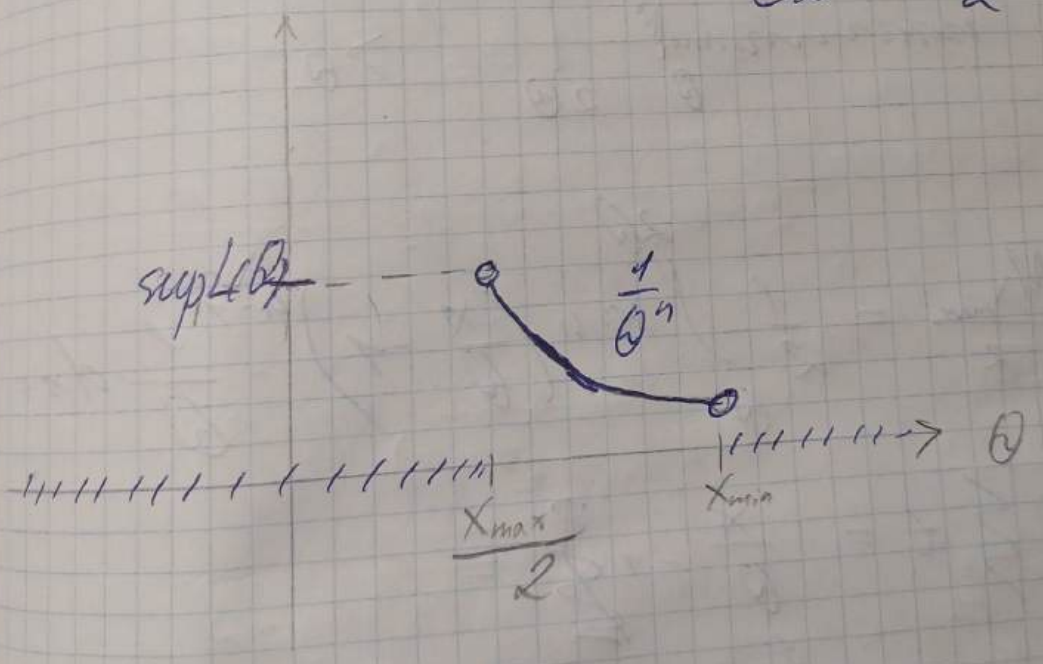
ОМП:

$$L(\theta) = \prod_{i=1}^{n} p(x,\theta) \Rightarrow$$

$$\Rightarrow L(\theta) = \frac{1}{\theta^n} \{(\theta < x_i < 2\theta)\}$$

$$\Rightarrow \begin{matrix} x_{max} < 2\theta \\ x_{min} > \theta \end{matrix} \Rightarrow \theta \in \left(\frac{x_{max}}{2}; x_{min}\right)$$

$$\Rightarrow \hat{\theta}_2 = \frac{x_{max}}{2}$$

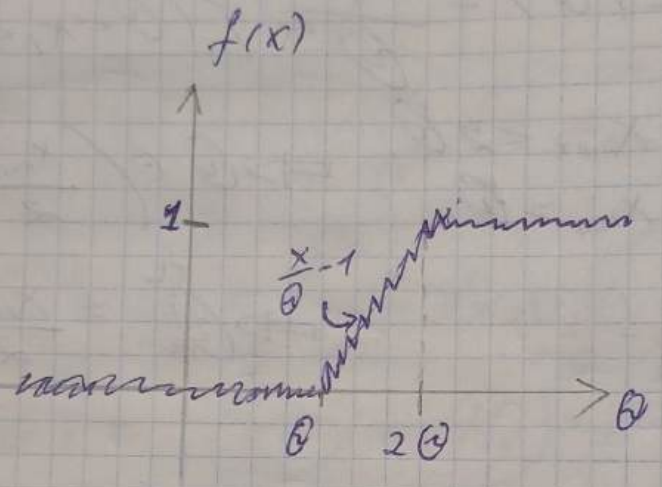


Нес.: $M\tilde{\theta}_2 = \theta$

$$M\tilde{\theta}_2 = M \cdot \frac{x_{max}}{2}$$

$$x_{max} \sim (f(x))^n \quad \psi(x)$$

$$\psi(x) = \psi'(x) = n\left(\frac{x}{\theta} - 1\right)^{n-1} \cdot \frac{1}{\theta} \{(\theta, 2\theta)\}$$




$$\frac{M x_{max}}{2} = \frac{1}{2} \int_{\theta}^{2\theta} x \cdot n\left(\frac{x}{\theta} - 1\right)^{n-1} \cdot \frac{1}{\theta} dx =$$

$$= \left\{ t = \frac{x}{\theta} - 1 \right\} =$$

$$= \frac{n\theta}{2} \int_0^1 (1+t)\, t^{n-1} dt =$$

$$= \frac{n\theta}{2} \int_0^1 (t^{n-1} + t^n) dt =$$

$$= \frac{n\theta}{2}\left(\frac{1}{n} + \frac{1}{n+1}\right) =$$

$$= \frac{n\theta}{2} \cdot \frac{(2n+1)}{n(n+1)} = \frac{\theta}{2} \cdot \frac{(2n+1)}{(n+1)} \Rightarrow$$

$$\Rightarrow \tilde{\theta}_2 \text{ — смещ.} \Rightarrow \tilde{\theta}_2^{\,*} = \frac{2(n+1)}{(2n+1)}\tilde{\theta}_2 =$$

$$= \left(\frac{n+1}{2n+1}\right) X_{max}$$

$$\Rightarrow M\tilde{\theta}_2^{\,*} = \theta \Rightarrow \tilde{\theta}_2^{\,*} \text{ — несмещ.}$$

<u>Сост.</u> $\tilde{\theta}_2$:

$$M\left[\frac{X_{max}}{2}\right]^2 = \frac{n\theta^2}{4}\int_0^1 (1+t)^2 t^{n-1}\,dt =$$

$$= \frac{n\theta^2}{4}\left(\frac{1}{n} + \frac{2}{n+1} + \frac{1}{n+2}\right) =$$

$$= \frac{\theta^2}{4} \cdot \frac{(n+1)(n+2) + 2n(n+2) + n(n+1)}{(n+1)(n+2)} =$$

$$= \frac{\theta^2(4n^2+8n+2)}{4(n+1)(n+2)}$$

$$D\tilde{\theta}_2 = M(\tilde{\theta}_2)^2 - (M\tilde{\theta}_2)^2 =$$

$$= \frac{\theta^2(4n^2+8n+2)}{4(n+1)(n+2)} - \frac{\theta^2(4n^2+4n+1)}{4(n+1)^2} =$$

$$= \frac{\theta^2}{4}\left(\frac{(4n^2+8n+2)(n+1) - (4n^2+4n+1)(n+2)}{(n+1)^2(n+2)}\right) =$$

$$= \frac{n\theta^2}{4(n+1)^2(n+2)} \xrightarrow[n\to\infty]{} 0$$

$\Rightarrow \tilde{\theta}_2$ – сост.

Сост. $\tilde{\theta}_2'$:

$$D\tilde{\theta}_2' = \frac{(2n+2)^2}{(2n+1)^2} D\tilde{\theta}_2 \to$$

$$\xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_2' \text{ – сост.}$$

Сравнение эффективности оценок

$$\tilde{\theta}_1 = \frac{2}{3}\bar{x}, \quad \tilde{\theta}_2' = \left(\frac{n+1}{2n+1}\right) x_{max}$$

$$D\tilde{\theta}_1 = \frac{\theta^2}{27n}; \quad D\tilde{\theta}_2' = \frac{n\theta^2}{(2n+1)^2(n+2)}$$

$$\frac{1}{27n} \;?\; \frac{n}{(2n+1)^2(n+1)}$$

$$27n^2 \;?\; (4n^2+4n+1)(n+2)$$

$$4n^3 - 15n^2 + 9n + 2 \;?\; 0$$

$$(n-1)(4n^2 - 11n - 2)$$

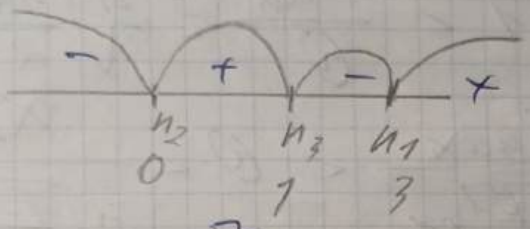
$4n^2 - 11n - 2 = 0$

$n_{1,2} = \frac{11}{8} \pm \frac{5}{8}\sqrt{17}$

$n_1 \simeq 3$ ; $n_2 \simeq 0$

округл. до целого значения

После $n = 3$: $D\tilde{\Theta}_1 \geq D\tilde{\Theta}_2$,




$\Longrightarrow \tilde{\Theta}_2$ эффективнее $\tilde{\Theta}_1$

Точный доверительный интервал для $\Theta$:

$f(\vec{X}_n, h) = \frac{X_{max}}{\Theta} - 1$

$P(f < t) = P(X_{max} < \Theta t + \Theta) =$ "

$= (f(\Theta t + \Theta))^n =$

$= \{ f(x, \Theta) = \left(\frac{x}{\Theta} - 1\right) \{(\Theta, 2\Theta)\} \} =$

$= (t + 1 - 1)^n \{\Theta, 2\Theta\} =$

$= t^n \{(\Theta, 2\Theta)\}$

$t_1$: $f(t_1) = \frac{1-\beta}{2}$

$t_2$: $f(t_2) = \frac{1+\beta}{2}$

$$\sqrt[n]{\frac{1-\beta}{2}} < \frac{X_{max}}{\theta} - 1 < \sqrt[n]{\frac{1+\beta}{2}}$$

$$P\left(\sqrt[n]{\frac{1-\beta}{2}} < \frac{X_{max}}{\theta} - 1 < \sqrt[n]{\frac{1+\beta}{2}}\right) = \beta$$

$$\sqrt[n]{\frac{1-\beta}{2}} + 1 < \frac{X_{max}}{\theta} < \sqrt[n]{\frac{1+\beta}{2}} + 1 \Rightarrow$$

$$\Rightarrow \frac{X_{max}}{\sqrt[n]{\frac{1+\beta}{2}} + 1} < \theta < \frac{X_{max}}{\sqrt[n]{\frac{1-\beta}{2}} + 1}$$

Асимптотический доверит. инт.:
модель нерегулярная $\Rightarrow$
$\Rightarrow$ нельзя применять
асимптот. оценку по ОМП.

<u>ОММ</u>: $\tilde{\theta}_1 = \frac{2}{3}\tilde{\alpha}_1 = g(\tilde{\alpha}_1)$:

$$\nabla g(\alpha) = \frac{2}{3}$$

$$\theta = g(\alpha): \quad K_{11} = \alpha_{1+1} - \alpha_1\alpha_1$$

$$\sqrt{n} \cdot \frac{g(\tilde{\alpha}) - g(\alpha)}{\sigma(\tilde{\alpha})} \rightsquigarrow N(0,1)$$

$$\sigma(\alpha))^2 = \nabla g(\alpha)^T K \nabla g(\alpha)$$

$$\sigma(\alpha)^2 = \frac{2}{3}(\alpha_2 - \alpha_1^2) \cdot \frac{2}{3} =$$

$$= \frac{4}{9}(\alpha_2 - \alpha_1^2)$$

$$\frac{\sqrt{n}\,(\hat{\theta}_1 - \theta)}{\frac{2}{3}\sqrt{\hat{\alpha}_2 - \hat{\alpha}_1^2}} \sim N(0,1):$$

$$U_{\frac{1-\beta}{2}} = t_1 \quad \Longrightarrow$$

$$U_{\frac{1+\beta}{2}} = t_2$$

$$\Longrightarrow P\left(t_1 < \frac{\sqrt{n}\,(\hat{\theta}_1 - \theta)}{\frac{2}{3}\sqrt{\hat{\alpha}_2 - \hat{\alpha}_1^2}} < t_2\right) = \beta$$

$$t_1 \cdot \frac{2}{3} \cdot \frac{\sqrt{\hat{\alpha}_2 - \hat{\alpha}_1^2}}{\sqrt{n}} < \hat{\theta}_1 - \theta <$$

$$< t_2 \cdot \frac{2}{3} \cdot \frac{\sqrt{\hat{\alpha}_2 - \hat{\alpha}_1^2}}{\sqrt{n}} \Longrightarrow$$

$$\Longrightarrow \hat{\theta}_1 - \frac{2}{3}\frac{\sqrt{\hat{\alpha}_2 - \hat{\alpha}_1^2}}{\sqrt{n}} \cdot t_1 <$$

$$< \theta < \hat{\theta}_1 - \frac{2}{3} \cdot \frac{\sqrt{\hat{\alpha}_2 - (\hat{\alpha}_1)^2}}{\sqrt{n}} + 1$$

$\boxed{N6}$

а) ОМП: $L(\theta) = \dfrac{(\theta-1)^n}{\prod_{i=1}^{n} x_i^{\theta}} = \prod_{i=1}^{n} p(x_i, \theta)$

$\ln L(\theta) = n \cdot \ln(\theta-1) - \theta \sum_{i=1}^{n} \ln(x_i) \Rightarrow$

$\Rightarrow \dfrac{\partial \ln(\theta)}{\partial \theta} = \dfrac{n}{\theta-1} - \sum_{i=1}^{n} \ln(x_i) = 0$

$\Rightarrow \dfrac{n}{\theta-1} = \sum_{i=1}^{n} \ln(x_i)$

$\dfrac{1}{n} \sum_{i=1}^{n} \ln(x_i) = \dfrac{1}{\theta-1} \Rightarrow$

$\Rightarrow \tilde{\theta} = 1 + \dfrac{1}{\frac{1}{n} \sum_{i=1}^{n} \ln(x_i)}$

$\dfrac{\partial^2 \ln L(\theta)}{\partial \theta^2} = \dfrac{-n}{(\theta-1)^2} < 0 \quad \forall \theta > 1$

$\Rightarrow$ лок. макс.

б) Усл. (1) об асимпт. св-вах ОМП:

1) $\Theta = (1, +\infty)$ – открытое

2) сильно рег.

$\to$ рег.

$$\to \frac{\partial^2}{(\partial\Theta)^2}\int_{-\infty}^{+\infty} d f(x,\Theta) =$$

$$= \int_{-\infty}^{+\infty} d\,\frac{\partial^2 f(x,\Theta)}{\partial\Theta^2}$$

$\to p(x,\Theta) \in C^2$

---

$p(x,\Theta)$ – непр. дифф. по $\Theta$ на $(1,+\infty)$

$$\int_1^{+\infty}\frac{\partial p(x,\Theta)}{\partial\Theta}dx = \int_1^{+\infty}\frac{\ln x - \Theta\ln x + 1}{x^{\Theta}}dx =$$

$$= \frac{x^{1-\Theta}}{1-\Theta}\Big|_1^{+\infty} + \int_1^{+\infty}\ln x\cdot d(x^{1-\Theta}) =$$

$$= \frac{1}{\Theta-1} - \frac{1}{\Theta-1} = 0$$

$$I(\Theta) = M\left[\left(\frac{\partial \ln p(x,\Theta)}{\partial\Theta}\right)^2\right] =$$

$$= \int_1^{+\infty} \left(\frac{\theta-1}{x^{\theta}}\right)\left(\frac{1}{\theta-1} - \ln x\right)^2 dx =$$

$$= \int_1^{+\infty} \left(\frac{1}{x^{\theta}(\theta-1)} - 2\cdot\frac{(\theta-1)\ln x}{x^{\theta}(\theta-1} + \right.$$

$$\left. + \frac{(\theta-1)(\ln x)^2}{x^{\theta}}\right) dx =$$

$$= \frac{1}{(\theta-1)^2} \underbrace{- 2\int_1^{+\infty} \frac{\ln x}{x^{\theta}} dx}_{I_1} +$$

$$\underbrace{+ \int_1^{+\infty} \frac{(\theta-1)(\ln x)^2}{x^{\theta}} dx}_{I_2} = \ldots$$

$=$

$$I_1 = -2\int_1^{+\infty} \ln x \cdot d\left(\frac{-1}{(\theta-1)x^{\theta-1}}\right) =$$

$$= -2\ln\left(x\cdot\frac{x^{1-\theta}}{\theta-1}\right)\Big|_1^{+\infty} +$$

$0$

$$+ 2\int_1^{+\infty} \frac{x^{-\theta}}{1-\theta}\,dx = 2\cdot\frac{1}{(1-\theta)^2}$$

$$I_2 = \int_1^{+\infty} \frac{(\theta-1)}{x^{\theta}} (\ln x)^2 dx =$$

$$= \int_1^{+\infty} (\ln x)^2 \, d(-x^{1-\theta}) =$$

$$= -x^{(1-\theta)}(\ln x)^2 \Big|_1^{+\infty} +$$

$$+ 2\int_1^{+\infty} \frac{x^{1-\theta}\cdot \ln x}{x}\cdot dx =$$

$$= 2\int_1^{+\infty} x^{-\theta} \ln x \cdot dx = \frac{2x^{1-\theta}}{1-\theta}\cdot \ln x \Big|_1^{+\infty} -$$

$$- 2\int_1^{+\infty} \frac{x^{-\theta}}{1-\theta}\,dx = -2\frac{1}{(1-\theta)^2}$$

$$\ldots = \frac{1}{(\theta-1)^2} + \frac{2}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} =$$

$$= \frac{1}{(\theta-1)^2}$$ – непр. на $[1,+\infty)$

$I(\theta) > 0$

$\Longrightarrow$ модель регулярна

$$\frac{\partial^2 p(x,\theta)}{\partial\theta^2} = \frac{-2\ln x + (\ln x)^2(\theta-1)}{x^\theta}$$

$$\frac{\partial^2}{\partial\theta^2}\int_1^{+\infty} p(x,\theta)\,dx = 0$$

$$\int_1^{+\infty}\left(\frac{-2\ln x}{x^\theta} + \frac{(\ln x)^2(\theta-1)}{x^\theta}\right)dx =$$

$$= \frac{2}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} = 0$$

$p(x,\theta)$ - 2ди непр. дифф.
на $\Theta = (1, +\infty)$

$\Rightarrow$ модель сильно регулярна

по Т ОМП - состоят., асимптот., несмещ., асимптот. нормальная, асимпт. эффективна

$p = 0{,}5$ (медиана)

$F(x_p) = p$

$f(x_{0,5}) = 0{,}5$

$$\int_1^{x} \left(\frac{\theta-1}{t^{\theta}}\right) dt = \frac{-1}{x^{\theta-1}} + 1 = \frac{1}{2}$$

$$x_{0,5} = \frac{1}{2^{\theta-1}} \Rightarrow g(\tilde{\theta}) = 2^{\frac{1}{\tilde{\theta}-1}}$$

$$g(\theta) = \frac{1}{2^{\theta-1}}$$

$$\nabla g(\tilde{\theta}) = 2^{\frac{1}{\tilde{\theta}-1}} \cdot \ln 2 \cdot \left(-\frac{1}{(\tilde{\theta}-1)^2}\right)$$

$$I^{-1}(\theta) = (\theta-1)^2 \Rightarrow$$

$$\Rightarrow (\sigma(\theta))^2 = \nabla^T g(\theta)\, I^{-1}(\theta) \cdot \nabla g(\theta)$$

$$(\sigma(\theta))^2 = \left(2 \cdot \frac{1}{\theta-1} \cdot \ln 2 \cdot \left(\frac{-1}{(\theta-1)^2}\right)\right)^2 \cdot$$

$$\cdot (\theta-1)^2$$

$$\Rightarrow \frac{\sqrt{n} \cdot \left(2^{\frac{1}{\theta-1}} - 2^{\frac{1}{\tilde{\theta}-1}}\right)}{2^{\frac{1}{\tilde{\theta}-1}} \cdot \ln 2 \cdot \frac{1}{(\tilde{\theta}-1)}} \rightsquigarrow N(0,1)$$

$$t_1 = U_{\frac{1-\beta}{2}} \; ; \; t_2 = U_{\frac{1+\beta}{2}}$$

$$P\left(t_1 < \frac{\sqrt{n}(\tilde{\theta}-1)\left(2\cdot\frac{1}{\tilde{\theta}-1} - 2\cdot\frac{1}{\tilde{\theta}-1}\right)}{2\cdot\frac{1}{\tilde{\theta}-1}\ln 2} < t_2\right) =$$

$$= \beta \Rightarrow$$

$$\Rightarrow \frac{2^{\frac{1}{\tilde{\theta}-1}}\cdot\ln 2}{\sqrt{n}(\tilde{\theta}^2-1)}\cdot t_1 < 2^{\frac{1}{\tilde{\theta}-1}} - x_{0,5} <$$

$$< \frac{2^{\frac{1}{\tilde{\theta}-1}}\cdot\ln 2}{\sqrt{n}\cdot(\tilde{\theta}-1)}\cdot t_2$$

$$2^{\frac{1}{\tilde{\theta}-1}} - \frac{2^{\frac{1}{\tilde{\theta}-1}}\cdot\ln 2}{\sqrt{n}(\tilde{\theta}-1)}\cdot t_2 < x_{0,5} <$$

$$< 2^{\frac{1}{\tilde{\theta}-1}} - \frac{2^{\frac{1}{\tilde{\theta}-1}}\cdot\ln 2}{\sqrt{n}\cdot(\tilde{\theta}-1)}\, t_1$$

c) $\sqrt{n}\cdot\dfrac{\tilde{\theta}-\theta}{\sigma(\tilde{\theta})} \rightsquigarrow N(0,1)$

$$g(\theta) = \theta \Rightarrow \nabla g(\theta) = 1$$

$$(\sigma(\theta))^2 = 1 \cdot (\theta - 1)^2 \cdot 1$$

$$\sqrt{n} \cdot \frac{\tilde{\theta} - \theta}{\theta - 1} \rightsquigarrow N(0, 1)$$

$$\tilde{\theta} = 1 + \frac{1}{\frac{1}{n} \sum_{i=1}^{n} \ln(x_i)}$$

$$\sqrt{n} \cdot \frac{1}{n} \cdot \sum_{i=1}^{n} \ln x_i (\tilde{\theta} - \theta) \rightsquigarrow N(0,1) \Rightarrow$$

$$\Rightarrow \tilde{\theta} - 1 = \frac{1}{\frac{1}{n} \sum_{i=1}^{n} \ln(x_i)} \cdot \frac{1}{\tilde{\theta} - 1} =$$

$$= \frac{1}{n} \sum_{i=1}^{n} \ln(x_i)$$

$$t_1 = U_{\frac{1-\beta}{2}} \; ; \; t_2 = U_{\frac{1+\beta}{2}}$$

$$P\left( t_1 < \frac{1}{\sqrt{n}} \cdot \sum_{i=1}^{n} \ln(x_i)(\tilde{\theta} - \theta) < t_2 \right) = \beta$$

$$\boxed{\tilde{\theta} - \frac{t_1}{\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \ln(x_i)} < \theta < \tilde{\theta} - \frac{t_1}{\frac{1}{\sqrt{n}} \cdot \sum_{i=1}^{n} \ln(x_i)}}$$

асимп. доверит. интервал по ОМП